Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

3-Plane Localizer - 3-Plane Localizer

IMAGING PARAMETERS	
Imaging Mode	*2D*
Pulse Sequence	*Spin Echo*
Imaging Options	*Seq, EDR, TRF, Fast, SS, ARC*
Phase	*2.00*

SCAN RANGE	
FOV	*24.0*
Slice Thickness	*10.00*
Slice Spacing	*5.0*
Overlap Locations	*0*

ACQ TIMING	
Freq	*256*
Phase	*192*
Freq DIR	*Unswap*
# of Acq. Before Pause	*0*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI	
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT	
Tag Type	*None*

TRICKS	
Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

SCAN TIMING	
TE	*80.0*
Number of Echoes	*1*
TR	*1000.0*
Receiver Bandwidth	*83.33*

IMAGE ENHANCE	
Filter Choice	*None*

USER CVS	
User CV Mask2	*0*

MULTI-PHASE	
Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION	
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST	
Contrast Yes/No	*No*

3-Plane Localizer - 3-Plane Localizer

Property of Wallace Tumor Institute

OTHERS

Protocol Notes

General Considerations:
-AIRx is optional software that provides automatic slice positioning for graphical prescriptions. Using pre-trained neural networks, AIRx calculates common Anatomical References from the 3-plane Localizer.
-To enable AIRx, acquire 3-plane Localizer with the following requirements:
-10cm anatomical coverage,
-80TE,
-192 Phase Matrix.
-Subsequent scans must be set to Oblique plane.
Important SAR Tips:
Improved Head SAR performance will be seen with during a brain exam if the SSFSE localizer calculation is lower than 3.2 W/kg
Keep the TR at 1000 msec in order to lower SAR estimations

AIR Recon DL Considerations:
To optimize existing 2D FSE protocols with AIR Recon DL:
1. Determine resolution needs. In order to improve resolution, consider the following adjustments:
a. Increase frequency matrix
b. Increase phase matrix
c. Decrease slice thickness
d. Decrease FOV
2. Increase bandwidth to minimize echo spacing and blurring.
3. Decrease scan time using the following parameters:
a. Decrease NEX
b. Increase the Acceleration factors with ARC or ASSET
c. Increase the Echo Train Length
Scan Parameter Tradeoffs:

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Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

PET Task

PATIENT POSITION	
Patient Entry	*Head First*
Patient Position	*Supine*

SCAN SETUP	
Scan Description	*PET BRAIN Static 30min AV1451*
Scan Type	*Static*
Gating Mode	*Off*
Cardiac State	*Off*
Start Location	*S 130.473*
End Location	*I 114.167*
Overlap	*0 Slices*
MRAC	*Yes*
Phantom AC Map	*None*
Nuclide	*18F*
Tracer	*F- -- Fluorine*
Number of PET Beds	*1*

CARDIAC/RESPIRATORY	
Binning Type	*NA*
Trigger Rejection	*NA*
Avg Trig/min	*NA*
% Dev. Allowed	*NA*

RECON2	
Recon Enabled	*Yes*
Series Description	*PET BRAIN MAC STATIC Reg*
DFOV	*30.000 cm*
Recon Method	*TOF*
Cardiac 3D Filter	*No*
Recon Start Location	*S 130.473*
Recon End Location	*I 114.167*
R/L Center	*-0.669 mm*
A/P Center	*{ pet_displayFieldOfViewAp pet_displayFieldOfViewRl } mm*
Z-Axis Filter	*Standard*
Filter Cutoff	*6.400 mm*
Subsets	*28*
Iterations	*6*
Matrix Size	*256 x 256*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*Yes*
Q.Clear Beta	*350.000*

PATIENT POSITION - REPLAY1	
Patient Entry	*Head First*
Patient Position	*Supine*

SCAN TIMING	
Pre-Scan Delay	*0*
Scan Time	*1800 sec*
Start on Count Rate	*0*
Stop on kcounts	*0*

SCAN DETAILS	
PET Bed 1 Scan Time	*300 sec*

RECON1	
Recon Enabled	*Yes*
Series Description	*PET BRAIN NAC STATIC Reg*
DFOV	*30.000 cm*
Recon Method	*TOF*
Cardiac 3D Filter	*No*
Recon Start Location	*S 130.473*
Recon End Location	*I 114.167*
R/L Center	*-0.669 mm*
A/P Center	*18.332 mm*
Z-Axis Filter	*Standard*
Filter Cutoff	*6.400 mm*
Subsets	*28*
Iterations	*6*
Matrix Size	*256 x 256*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*No*
Q.Clear Beta	*350.000*

RECON3	
Recon Enabled	*Yes*
Series Description	*PET BRAIN MAC STATIC Reg ZTE*
DFOV	*30.000 cm*
Recon Method	*TOF*
Cardiac 3D Filter	*No*
Recon Start Location	*S 130.473*
Recon End Location	*I 114.167*
R/L Center	*-0.669 mm*
A/P Center	*{ pet_displayFieldOfViewAp pet_displayFieldOfViewRl } mm*
Z-Axis Filter	*Standard*
Filter Cutoff	*6.400 mm*
Subsets	*28*
Iterations	*6*
Matrix Size	*256 x 256*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*Yes*
Q.Clear Beta	*350.000*

PET Task

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SCAN SETUP - REPLAY1	
Scan Description	*6 Ph PET BRAIN F-18 AV1451*
Scan Type	*Dynamic*
Gating Mode	*Off*
Cardiac State	*Off*
Start Location	*S 130.473*
End Location	*I 114.167*
Overlap	*0 Slices*
MRAC	*Yes*
Phantom AC Map	*None*
Nuclide	*F- -- Fluorine*
Tracer	*18F*
Number of PET Beds	*1*

RECON1 - REPLAY1	
Recon Enabled	*Yes*
Series Description	*6 Ph PET BRAIN MAC TOF REG*
DFOV	*30.000 cm*
Recon Method	*TOF*
Cardiac 3D Filter	*No*
Recon Start Location	*S 130.473*
Recon End Location	*I 114.167*
R/L Center	*7.500 mm*
A/P Center	*11.500 mm*
Z-Axis Filter	*Standard*
Filter Cutoff	*6.400 mm*
Subsets	*28*
Iterations	*6*
Matrix Size	*256 x 256*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*Yes*
Q.Clear Beta	*350.000*

RECON3 - REPLAY1	
Recon Enabled	*No*

SCAN TIMING - REPLAY1	
Pre-Scan Delay	*0*
Scan Time	*1800 sec*
Start on Count Rate	*0*
Stop on kcounts	*0*

CARDIAC/RESPIRATORY - REPLAY1	
Binning Type	*NA*
Trigger Rejection	*NA*
Avg Trig/min	*NA*
% Dev. Allowed	*NA*

RECON2 - REPLAY1	
Recon Enabled	*No*

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

MRAC 1 - PET BRAIN Static 30min AV1451

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*MRAC 1 - PET BRAIN Static 30min AV1451*

SCAN TIMING

Flip Angle	*5.0*
TE	*Min Full*
Number of Echoes	*2*
Receiver Bandwidth	*166.67*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV6	*1.00*
User CV30	*1.00*
User CV34	*70.00*
User CV Mask2	*8*

MULTI-PHASE

Seperate Series	*0*
Trigger Delay without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*LAVA*
Imaging Options	*EDR, Fast, Flex*
IDEAL	*15*

SCAN RANGE

FOV	*50.0*
Slice Thickness	*2.60*
Location per Slab	*120*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*256*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*0.75*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

MRAC 1 - PET BRAIN Static 30min AV1451

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

ZTE-MRAC 32Ch NOVA 1 - PET BRAIN Static 30min AV1451

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*ZTE-MRAC 32Ch NOVA 1 - PET BRAIN Static 30min AV1451*

SCAN TIMING

Flip Angle	*0.8*
Number of Echoes	*1*
Receiver Bandwidth	*62.50*
Recovery Time	*0*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*Gradient Echo*
Imaging Options	*EDR, Fast, SqP*
PSD Name	*3dradial*

SCAN RANGE

FOV	*26.4*
Slice Thickness	*2.40*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*110*
Freq DIR	*R/L*
NEX	*4.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*
Spokes Per Segment	*512*

USER CVS

User CV Mask2	*0*

MULTI-PHASE

Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

ZTE-MRAC 32Ch NOVA 1 - PET BRAIN Static 30min AV1451

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

3D Sag T1 BRAVO 1mm Isotropic HyperSense

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*3D Sag T1 BRAVO 1mm Isotropic HyperSense*

SCAN TIMING

Flip Angle	*12.0*
Number of Echoes	*1*
TI	*450*
Receiver Bandwidth	*31.25*
Recovery Time	*0*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV6	*1.00*
User CV30	*1.00*
User CV Mask2	*0*

MULTI-PHASE

Seperate Series	*0*
Trigger Delay without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
Multi b-values	*1000.0;*
Multi NEX Values	*1.0;*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*BRAVO*
Imaging Options	*EDR, Fast, ARC, HS, IrP*
Phase	*2.00*
Slice	*1.00*
HyperSense	*1.20*

SCAN RANGE

FOV	*25.6*
Slice Thickness	*1.00*
Location per Slab	*256*
Overlap Locations	*0*

ACQ TIMING

Freq	*256*
Phase	*256*
Freq DIR	*S/I*
Fat Shift DIR	*Normal (S)*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*2*

OTHERS

Protocol Notes	*General Considerations: -3D imaging can improve the slice plane resolution compared to 2D imaging -Prescribe the slices as shown*

3D Sag T1 BRAVO 1mm Isotropic HyperSense

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

Ax Task Run1 HB2x3 2200TR 2_1iso

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*Ax Task Run1 HB2x3 2200TR 2_1iso*

SCAN TIMING

Flip Angle	*90.0*
TE	*30.0*
Number of Echoes	*1*
TR	*2200.0*
Number of Shots	*1*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

Initial State	*Control*
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*This paradigm design is Resting State. First 10 seconds are dummy scans to get to stady state. At 8:20 the Resting State acquisition starts. Have the patient relax and not think of anything specific. The patient can either keep eyes open, keep eyes closed, or fixate on a specifiied location. Optional GE Brainwave Hardware includes pre-loaded library of patient instructions.*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*Gradient Echo*
Imaging Options	*EDR, MPh, EPI, ARC, HB*
PSD Name	*epialtoff*
Phase	*2.00*
HyperBand Slice	*3*

SCAN RANGE

FOV	*27.5*
Slice Thickness	*2.10*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*72*

ACQ TIMING

Freq	*128*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

USER CVS

User CV0	*1.00*
User CV11	*1.00*
User CV19	*1.00*
User CV Mask2	*0*

MULTI-PHASE

Slice per Location	*157*
Phase Acquisition Order	*Interleaved*
Delay after Acquisition	*Minimum*
Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*

DIFFUSION

Optimized TE	*Yes*
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

Ax Task Run1 HB2x3 2200TR 2_1iso

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

ALT ON 1 Spin Echo EPI

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*ALT ON 1 Spin Echo EPI*
SCAN TIMING	
TE	*30.0*
Number of Echoes	*1*
TR	*2200.0*
Number of Shots	*1*
IMAGE ENHANCE	
Filter Choice	*None*
GATING/TRIGGER	
Pause After Navigator Prescan	*0*
FMRI	
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*
SAT	
Tag Type	*None*
Fat/Water Saturation	*Fat*
TRICKS	
Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

IMAGING PARAMETERS	
Imaging Mode	*2D*
Pulse Sequence	*Spin Echo*
Imaging Options	*MPh, EPI, Asset*
PSD Name	*epialt*
Phase	*2.00*
SCAN RANGE	
FOV	*27.5*
Slice Thickness	*2.10*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*72*
ACQ TIMING	
Freq	*128*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*
USER CVS	
User CV0	*1.00*
User CV11	*1.00*
User CV Mask2	*0*
MULTI-PHASE	
Slice per Location	*3*
Phase Acquisition Order	*Interleaved*
Delay after Acquisition	*Minimum*
Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
DIFFUSION	
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*
CONTRAST	
Contrast Yes/No	*No*

ALT ON 1 Spin Echo EPI

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

ALT OFF 1 Spin Echo EPI

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*ALT OFF 1 Spin Echo EPI*
SCAN TIMING	
TE	*30.0*
Number of Echoes	*1*
TR	*2200.0*
Number of Shots	*1*
IMAGE ENHANCE	
Filter Choice	*None*
GATING/TRIGGER	
Pause After Navigator Prescan	*0*
FMRI	
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*
SAT	
Tag Type	*None*
Fat/Water Saturation	*Fat*
TRICKS	
Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

IMAGING PARAMETERS	
Imaging Mode	*2D*
Pulse Sequence	*Spin Echo*
Imaging Options	*MPh, EPI, Asset*
PSD Name	*epialtoff*
Phase	*2.00*
SCAN RANGE	
FOV	*27.5*
Slice Thickness	*2.10*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*72*
ACQ TIMING	
Freq	*128*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*
USER CVS	
User CV0	*1.00*
User CV11	*1.00*
User CV Mask2	*0*
MULTI-PHASE	
Slice per Location	*3*
Phase Acquisition Order	*Interleaved*
Delay after Acquisition	*Minimum*
Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
DIFFUSION	
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*
CONTRAST	
Contrast Yes/No	*No*

ALT OFF 1 Spin Echo EPI

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

Ax Task Run2 HB2x3 2200TR 2_1iso

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*Ax Task Run2 HB2x3 2200TR 2_1iso*

SCAN TIMING

Flip Angle	*90.0*
TE	*30.0*
Number of Echoes	*1*
TR	*2200.0*
Number of Shots	*1*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

Initial State	*Control*
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*This paradigm design is Resting State. First 10 seconds are dummy scans to get to stady state. At 8:20 the Resting State acquisition starts. Have the patient relax and not think of anything specific. The patient can either keep eyes open, keep eyes closed, or fixate on a specifiied location. Optional GE Brainwave Hardware includes pre-loaded library of patient instructions.*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*Gradient Echo*
Imaging Options	*EDR, MPh, EPI, ARC, HB*
PSD Name	*epialtoff*
Phase	*2.00*
HyperBand Slice	*3*

SCAN RANGE

FOV	*27.5*
Slice Thickness	*2.10*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*72*

ACQ TIMING

Freq	*128*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

USER CVS

User CV0	*1.00*
User CV11	*1.00*
User CV19	*1.00*
User CV Mask2	*0*

MULTI-PHASE

Slice per Location	*157*
Phase Acquisition Order	*Interleaved*
Delay after Acquisition	*Minimum*
Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*

DIFFUSION

Optimized TE	*Yes*
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

Ax Task Run2 HB2x3 2200TR 2_1iso

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

Ax Task Run3 HB2x3 2200TR 2_1iso

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*Ax Task Run3 HB2x3 2200TR 2_1iso*

SCAN TIMING

Flip Angle	*90.0*
TE	*30.0*
Number of Echoes	*1*
TR	*2200.0*
Number of Shots	*1*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

Initial State	*Control*
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*This paradigm design is Resting State. First 10 seconds are dummy scans to get to stady state. At 8:20 the Resting State acquisition starts. Have the patient relax and not think of anything specific. The patient can either keep eyes open, keep eyes closed, or fixate on a specifiied location. Optional GE Brainwave Hardware includes pre-loaded library of patient instructions.*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*Gradient Echo*
Imaging Options	*EDR, MPh, EPI, ARC, HB*
PSD Name	*epialtoff*
Phase	*2.00*
HyperBand Slice	*3*

SCAN RANGE

FOV	*27.5*
Slice Thickness	*2.10*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*72*

ACQ TIMING

Freq	*128*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

USER CVS

User CV0	*1.00*
User CV11	*1.00*
User CV19	*1.00*
User CV Mask2	*0*

MULTI-PHASE

Slice per Location	*157*
Phase Acquisition Order	*Interleaved*
Delay after Acquisition	*Minimum*
Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*

DIFFUSION

Optimized TE	*Yes*
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

Ax Task Run3 HB2x3 2200TR 2_1iso

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

ALT ON 2 Spin Echo EPI

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*ALT ON 2 Spin Echo EPI*

SCAN TIMING

TE	*30.0*
Number of Echoes	*1*
TR	*2200.0*
Number of Shots	*1*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*
Fat/Water Saturation	*Fat*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*Spin Echo*
Imaging Options	*MPh, EPI, Asset*
PSD Name	*epialt*
Phase	*2.00*

SCAN RANGE

FOV	*27.5*
Slice Thickness	*2.10*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*72*

ACQ TIMING

Freq	*128*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

USER CVS

User CV0	*1.00*
User CV11	*1.00*
User CV Mask2	*0*

MULTI-PHASE

Slice per Location	*3*
Phase Acquisition Order	*Interleaved*
Delay after Acquisition	*Minimum*
Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

ALT ON 2 Spin Echo EPI

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

ALT OFF 2 Spin Echo EPI

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*ALT OFF 2 Spin Echo EPI*

SCAN TIMING

TE	*30.0*
Number of Echoes	*1*
TR	*2200.0*
Number of Shots	*1*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*
Fat/Water Saturation	*Fat*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*Spin Echo*
Imaging Options	*MPh, EPI, Asset*
PSD Name	*epialtoff*
Phase	*2.00*

SCAN RANGE

FOV	*27.5*
Slice Thickness	*2.10*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*72*

ACQ TIMING

Freq	*128*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

USER CVS

User CV0	*1.00*
User CV11	*1.00*
User CV Mask2	*0*

MULTI-PHASE

Slice per Location	*3*
Phase Acquisition Order	*Interleaved*
Delay after Acquisition	*Minimum*
Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

ALT OFF 2 Spin Echo EPI

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

Ax Rest 1 HB2x3 2200TR 2_1iso

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*Ax Rest 1 HB2x3 2200TR 2_1iso*

SCAN TIMING

Flip Angle	*90.0*
TE	*30.0*
Number of Echoes	*1*
TR	*2200.0*
Number of Shots	*1*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

Initial State	*Control*
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*This paradigm design is Resting State. First 10 seconds are dummy scans to get to stady state. At 8:20 the Resting State acquisition starts. Have the patient relax and not think of anything specific. The patient can either keep eyes open, keep eyes closed, or fixate on a specifiied location. Optional GE Brainwave Hardware includes pre-loaded library of patient instructions.*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*Gradient Echo*
Imaging Options	*EDR, MPh, EPI, ARC, HB*
PSD Name	*epialtoff*
Phase	*2.00*
HyperBand Slice	*3*

SCAN RANGE

FOV	*27.5*
Slice Thickness	*2.10*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*72*

ACQ TIMING

Freq	*128*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

USER CVS

User CV0	*1.00*
User CV11	*1.00*
User CV19	*1.00*
User CV Mask2	*0*

MULTI-PHASE

Slice per Location	*200*
Phase Acquisition Order	*Interleaved*
Delay after Acquisition	*Minimum*
Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*

DIFFUSION

Optimized TE	*Yes*
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

Ax Rest 1 HB2x3 2200TR 2_1iso

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

SIMULTANEOUS MR - Bed 1

SAG 3D T2

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*SAG 3D T2*

SCAN TIMING

TE	*Maximum*
Number of Echoes	*1*
TR	*3200.0*
Echo Train Length	*100*
Receiver Bandwidth	*62.50*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV29	*1.00*
User CV30	*1.00*
User CV32	*1.00*
User CV34	*100.00*
User CV Mask2	*522*

MULTI-PHASE

Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*Cube*
Imaging Options	*EDR, Fast, FR, ARC*
Phase	*2.00*
Slice	*2.00*

SCAN RANGE

FOV	*25.6*
Slice Thickness	*1.00*
Location per Slab	*200*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*256*
Phase	*256*
Freq DIR	*S/I*
Fat Shift DIR	*Normal (S)*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*2*

SAG 3D T2

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: VISLAB ADRC AV 1451 WITH TASK 06/11/26

OTHERS	
Protocol Notes	*Cube 3D Imaging Considerations:* *-Use Cube to acquire high resolution imaging in all three dimensions with the goal of acquiring isotropic voxels (all voxel dimensions are equl-height, width, and depth). Consider this information when modifying Cube T2 or Cube T2 FLAIR scan parameters.* *-Scan selections: 3D Mode, Fast Spine Echo family, Cube or Cube T2 FLAIR pulse, Extended Dynamic Range and IR Prepared Imaging Options.* *-Typically use Cube to acquire brain, C-spine, pelvis and keee images.* *-Cube brain scans acquired with User CV Cube Enhance set to 3 (Brain T1) can produce 3D isotropic images with reduced artifacts in comparison to a traditional 2D Spin Echo scan. However, residual flow artifacts and ringing artifact may still be present.* *-When acquring a Cube pelvis scan, it is recommended you acquire a sagittal or coronal plane. Acquiring an axial plane may result in annefact artifact.* *-From the scan parameters Details screen, consider selecting Excitation Mode: Non-Selective.* *-From the scan parameters Details screen, consider selecting Excitation Mode: Focus to redue phase alias artifact when a small FOV is used.*

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